

Figure 1: Limitations of the standard secant condition. (a) Function values and gradients at x_k and x_{k+1} are already known. (b) The ideal quadratic model constructed from the exact Hessian fits well around the new point x_{k+1} . (c) The standard secant update may fail to capture curvature adequately.

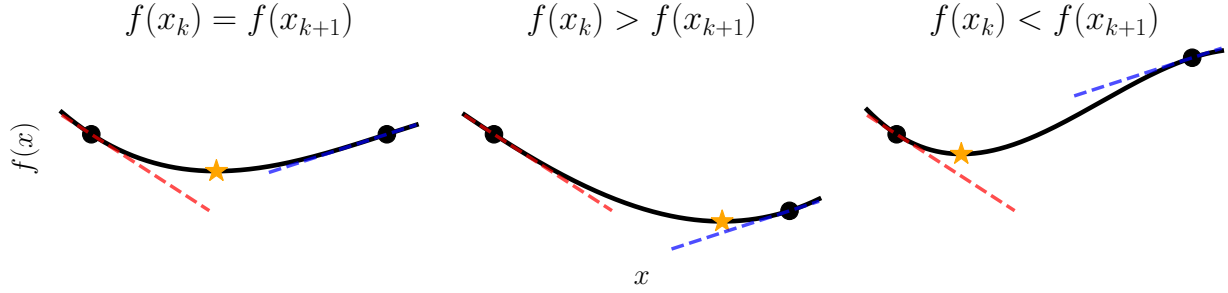


Figure 2: Interpolation with identical gradients at x_k and x_{k+1} but different function values. This leads to distinct interpolants, highlighting the importance of incorporating function value information in Hessian approximation.

1 Modified Secant Condition

This section explains the modified secant condition, a modification of the standard secant condition used in quasi-Newton methods. The modified secant condition incorporates function value information in addition to gradient information, leading to more accurate Hessian approximations.

1.1 Modified Secant Condition

Although the secant condition is widely used in many quasi-Newton methods, it can fail to capture the curvature of the objective function accurately because it only utilizes gradient information. We illustrate this limitation in Figure 1. In Figure 1a, we have two points x_k and x_{k+1} with their corresponding function values and gradients. In Figure 1b, we see that the ideal quadratic model constructed from the exact Hessian fits well around the new point x_{k+1} , leading to a better convergence behavior. However, in Figure 1c, the standard secant update only matches the gradients at these points, neglecting the function value at x_k . It can severely misestimate the curvature, leading to a poor approximation of the true objective function.

We can overcome this limitation by utilizing the function values. The basic idea is illustrated in Figure 2. Even when the gradients are the same at two points x_k and x_{k+1} , natural interpolation of function values differs depending on the function values $f(x_k)$ and $f(x_{k+1})$. This observation motivates the modified secant condition, which incorporates function value information to improve Hessian approximation.

In the following, we present two well-known modified secant conditions.

1.1.1 Function-Value-Matching Modified Secant Condition

The first modification enforces the quadratic model to match the function value at the previous point [1, 4, 6]. Let us consider an alternate model with a different approximate Hessian B_{k+1}^F :

$$m_{k+1}^F(x) = f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2} (x - x_{k+1})^\top B_{k+1}^F (x - x_{k+1}).$$

We require this model to satisfy

$$m_{k+1}^F(x_k) = f(x_k),$$

which enforces that the function value at the previous point is correctly modeled.

Let us derive the corresponding modified secant condition for B_{k+1}^F . Substituting the definition of m_{k+1}^F and using $s_k = x_{k+1} - x_k$, the condition becomes

$$\begin{aligned} f(x_k) &= f(x_{k+1}) + \nabla f(x_{k+1})^\top (x_k - x_{k+1}) + \frac{1}{2} (x_k - x_{k+1})^\top B_{k+1}^F (x_k - x_{k+1}) \\ &= f(x_{k+1}) - \nabla f(x_{k+1})^\top s_k + \frac{1}{2} s_k^\top B_{k+1}^F s_k. \end{aligned}$$

Now, assume that the updated matrix has the form

$$B_{k+1}^F s_k = y_k - \sigma_k^F s_k.$$

Here, $\sigma_k^F \in \mathbb{R}$ is a scalar to be determined. Substituting this equation and using $s_k^\top B_{k+1}^F s_k = s_k^\top y_k - \sigma_k^F \|s_k\|^2$ gives

$$f(x_k) = f(x_{k+1}) - \nabla f(x_{k+1})^\top s_k + \frac{1}{2} s_k^\top y_k - \frac{\sigma_k^F}{2} \|s_k\|^2.$$

Solving for σ_k^F and using $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$, we obtain

$$\begin{aligned} \sigma_k^F &= \frac{2(f(x_{k+1}) - f(x_k)) - (2\nabla f(x_{k+1}) - y_k)^\top s_k}{\|s_k\|^2} \\ &= \frac{2(f(x_{k+1}) - f(x_k)) - (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k}{\|s_k\|^2}. \end{aligned}$$

Therefore, the modified secant condition becomes

$$B_{k+1}^F s_k = y_k + \frac{2(f(x_k) - f(x_{k+1})) + (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k}{\|s_k\|^2} s_k.$$

To preserve the possibility of maintaining positive definiteness under a BFGS-type update when $s_k^\top y_k > 0$, we can modify the formula by taking the maximum with zero in the numerator.

$$B_{k+1}^{F'} s_k = y_k + \frac{\max(0, 2(f(x_k) - f(x_{k+1})) + (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k)}{\|s_k\|^2} s_k.$$

This is the function-value-matching modified secant condition. Unlike the standard secant condition, this formulation ensures that the quadratic model matches not only the function value at the previous point but also the gradients at both points. Refer to Figure 3 for an illustration.

1.1.2 Cubic-Augmented Modified Secant Condition

The second modification introduces a cubic term to the model, allowing simultaneous satisfaction of both function value and gradient matching at the previous point [5, 7, 8]. Let $T_{k+1} \in \mathbb{R}^{n \times n \times n}$ be the third-order derivative tensor of f at x_{k+1} such that

$$s_k^\top (T_{k+1} s_k) s_k = \sum_{i,j,l=1}^n \partial_{x_i x_j x_l} f(x_{k+1}) s_k^{(i)} s_k^{(j)} s_k^{(l)},$$

where $\partial_{x_i x_j x_l} f$ denotes the third derivative of f with respect to x_i , x_j , and x_l , and $s_k^{(i)}$ is the i -th component of the vector s_k . We introduced this tensor solely for analysis purposes, and it will be eliminated from the final formula.

By incorporating this tensor term, we can define the following cubic-augmented model:

$$m_{k+1}^C(x) = f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2} (x - x_{k+1})^\top B_{k+1}^C (x - x_{k+1})$$

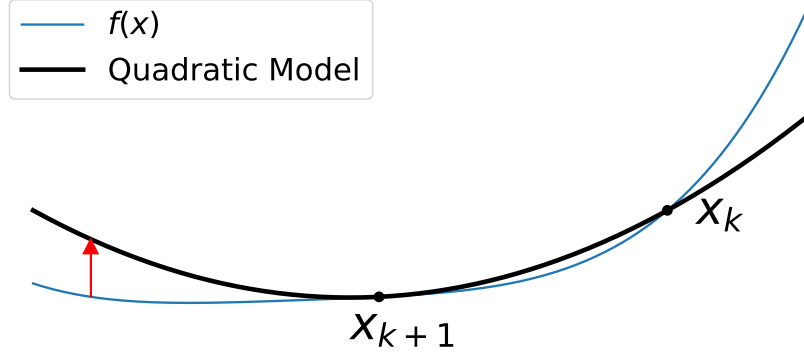


Figure 3: Function-value-matching modified secant equation. The modified secant equation $B_k^F s_k = y_k + \sigma_k^F s_k$ constructs a quadratic model that satisfies the function value condition $m_k^F(x_{k-1}) = f(x_{k-1})$ at the previous point. This differs from the standard secant equation, which only matches gradients but not function values.

$$+ \frac{1}{6}(x - x_{k+1})^\top (T_{k+1}(x - x_{k+1}))(x - x_{k+1}).$$

With this model, we can enforce both function value and gradient matching at the previous point x_k . Specifically, we require

$$\begin{cases} m_{k+1}^C(x_k) = f(x_k) \\ \nabla m_{k+1}^C(x_k) = \nabla f(x_k) \end{cases}$$

By substituting the definition of m_{k+1}^C and using $s_k = x_{k+1} - x_k$, we can rewrite these conditions as

$$\begin{aligned} f(x_k) &= f(x_{k+1}) - s_k^\top \nabla f(x_{k+1}) + \frac{1}{2} s_k^\top B_{k+1}^C s_k - \frac{1}{6} s_k^\top (T_{k+1} s_k) s_k, \\ \nabla f(x_k) &= \nabla f(x_{k+1}) - B_{k+1}^C s_k + \frac{1}{2} (T_{k+1} s_k) s_k. \end{aligned}$$

Rearranging and multiplying both sides by 3 for the first equation, and taking the inner product with s_k for the second equation, we have

$$\begin{aligned} 3(f(x_k) - f(x_{k+1})) + 3s_k^\top \nabla f(x_{k+1}) &= \frac{3}{2} s_k^\top B_{k+1}^C s_k - \frac{1}{2} s_k^\top (T_{k+1} s_k) s_k, \\ -s_k^\top y_k &= -s_k^\top B_{k+1}^C s_k + \frac{1}{2} s_k^\top (T_{k+1} s_k) s_k. \end{aligned}$$

By summing up and using $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$, we eliminate the tensor term and obtain the scalar identity

$$3(f(x_k) - f(x_{k+1})) + \frac{3}{2} s_k^\top (\nabla f(x_{k+1}) + \nabla f(x_k)) + \frac{1}{2} s_k^\top y_k = \frac{1}{2} s_k^\top B_{k+1}^C s_k.$$

Assume again that $B_{k+1}^C s_k$ is a linear combination of y_k and s_k , i.e., $B_{k+1}^C s_k = y_k + \sigma_k^C s_k$ for some scalar σ_k^C . Then, the previous equation yields

$$\sigma_k^C = \frac{6(f(x_k) - f(x_{k+1})) + 3s_k^\top (\nabla f(x_k) + \nabla f(x_{k+1}))}{\|s_k\|^2}.$$

Thus, the modified secant condition becomes

$$B_{k+1}^C s_k = y_k + \frac{6(f(x_k) - f(x_{k+1})) + 3s_k^\top (\nabla f(x_k) + \nabla f(x_{k+1}))}{\|s_k\|^2} s_k.$$

This is a cubic-augmented modified secant condition. Unlike the function-value-matching quadratic modification, this formulation allows simultaneous satisfaction of both the function value and gradient conditions at the previous point through the introduction of the tensor term. Refer to Figure 4 for an illustration.

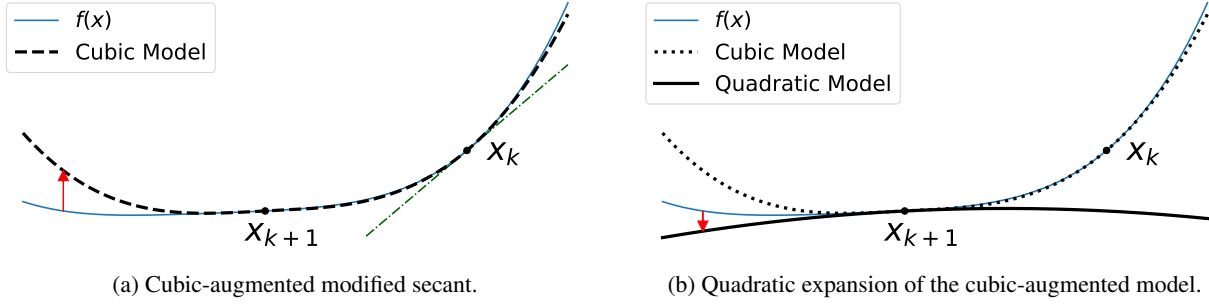


Figure 4: Cubic-augmented modified secant equation. In Figure 4a, by incorporating a cubic term $\eta \|x - x_k\|^3$ in the model, the modified secant equation $B_k^C s_k = y_k + \sigma_k^C s_k$ enables simultaneous satisfaction of both conditions at the previous point. In Figure 4b, while the cubic model satisfies both function value and gradient conditions, its underlying quadratic component may be indefinite or negative definite.

1.2 Other Curvature Storage Methods

For storing the curvature information, there are several topics. Agg-BFGS [2] is an alternative approach to managing curvature information by aggregating data rather than discarding the oldest information and adding the newest. Multi-Secant [3] extends the secant condition framework by maintaining multiple pairs of step and gradient difference vectors. In the standard formulation, we define

$$s_i = x_{i+1} - x_i, \quad y_i = \nabla f(x_{i+1}) - \nabla f(x_i). \quad (i = k - m, \dots, k)$$

An alternative anchored formulation centers all vectors at the most recent iterate:

$$s_i = x_{k+1} - x_i, \quad y_i = \nabla f(x_{k+1}) - \nabla f(x_i). \quad (i = k - m, \dots, k)$$

This anchored approach provides a different perspective on utilizing historical information for improved Hessian approximation.

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